

# RUI QIN

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## 🎓 EDUCATION

**Zhejiang University**, Hangzhou, China 2023 – Present

*PhD student* in Pharmacy, expected June 2028

*M.S. student* in AI Pharmacy (2023 – 2025); transferred to PhD track

**Sun Yat-Sen University**, Guangzhou, China 2019 – 2023

*B.S.* in Pharmaceutical Sciences

## 🔬 RESEARCH EXPERIENCE

**Large Language Model Systems for Drug Design** 2023 – Present

*PhD Thesis Project* Zhejiang University

- Developed chemical language model frameworks based on autoregressive and diffusion-based architectures as foundational model for multi-task drug design scenarios.
- Applied chemical language models to task-specific drug discovery problems, including molecule generation, optimization, and property prediction.

**Application of Molecular Generative Models in Drug Design** 2021 – Present

*Research Project* Zhejiang University & Sun Yat-Sen University

- Applied molecular generative models to practical drug design scenarios and evaluated generated molecules.
- Developed benchmarks for molecular generation using virtual screening, cheminformatics evaluation, and drug-likeness analysis.

## 📄 SELECTED PUBLICATIONS

1. **Rui Qin\***, Zijie Chen\*, Yurong Li, et al. *Revisiting Target-Aware de novo Molecular Generation with TarPass: Between Rational Design and Texas Sharpshooter*. **Advanced Science**, 2026.
2. Jike Wang\*, **Rui Qin\***, Mingyang Wang\*, et al. *Token-Mol 1.0: tokenized drug design with large language models*. **Nature Communications**, 2025.
3. Jike Wang\*, Hao Luo\*, **Rui Qin\***, et al. *3DSMILES-GPT: 3D molecular pocket-based generation with token-only large language model*. **Chemical Science**, 2025.
4. **Rui Qin**, Hao Zhang, Weifeng Huang, et al. *Deep learning-based design and screening of benzimidazole-pyrazine derivatives as adenosine A<sub>2B</sub> receptor antagonists*. **Journal of Biomolecular Structure and Dynamics**, 2023.

## 🏆 SELECTED HONORS AND FUNDING

### Funding:

*Scientific Research Fund of Zhejiang Provincial Education Department*, Principal Investigator 2025

### Honors and Awards:

*Graduate Academic Excellence Award*, Zhejiang University 2025

*Wong Lo Kat Scholarship*, School of Pharmaceutical Sciences, Sun Yat-Sen University 2023

*Academic Innovation Award*, Sun Yat-Sen University 2022

*Second-Class Academic Scholarship*, Sun Yat-Sen University 2022

## ⚙️ SKILLS

- Programming: Python, PyTorch, Linux
- Machine Learning: Large Language Models, Representation Learning

- Drug Design: Molecular Generation, Virtual Screening, Docking (Schrödinger, Vina), Cheminformatics, Molecular Dynamic (Gromacs, Amber)
- Toolkits: RDKit, Huggingface Transformers, PyTorch Geometric, OpenBabel, PyMOL